

Fraud that learns needs a defense that **learns back.**

A closed-loop adversarial lab for GenAI-era payment fraud. Discover emerging attacks, simulate them at fidelity across card, real-time and agentic rails, and harden a detector on exactly what breaks through.

— THE PROBLEM

Static defenses decay the moment fraud changes shape

450%+

dark-web "AI agent" chatter

Visa, H1 2026. A brand-new, machine-speed attack surface as agentic checkout scales.

524,121

mule accounts flagged

India, March 2026. Real-time rails launder funds in seconds (NPCI MuleHunter.AI).

25%

rise in malicious bot payments

Visa, six months. Delegated agent tokens transact with no cardholder step-up.

Evaluating one classifier on a fixed dataset measures fit to yesterday's fraud, not robustness to an adaptive adversary. The challenge asks for a loop. I built one.

— WHAT IT DOES

Build the attacks, then build the defense that beats them - as one loop

1 · Identify

Catalogue how GenAI attacks payments - fake identities, deepfake scam calls, hijacked shopping agents, mule rings - as a structured threat map.

2 · Generate

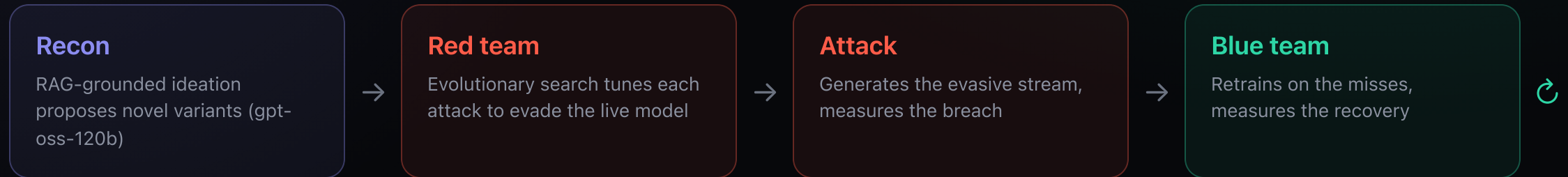
Turn each attack into realistic transactions on card, real-time and agentic rails. An AI red team mutates them to slip past my own detector.

3 · Defend

Train a detector on that data, measure it honestly, and retrain on whatever gets through. Repeat until the attacker can't win.

— ORCHESTRATION

A live multi-agent system (LangGraph)



A compiled StateGraph cycles recon → red_team → attack → blue_team until the round budget is spent. This report was generated by the graph.

```
$ baseline: detector recall 72.9% @ thr=1.000
$ [round 1] red_team: evolved evasive parameters for 9 vectors
$ [round 1] recon: 9 RAG-grounded variants proposed (groq:openai/gpt-oss-120b)
$ [round 1] attack: evasive stream breaches to 19.2% recall
$ [round 1] blue_team: retrained, recovers to 82.9% recall
$ [round 2] red_team: evolved evasive parameters for 9 vectors
```

— GENERATE

Fidelity is the point

Realistic by construction. 237,905 events, 1.39% fraud, across card-not-present, card-present, real-time A2A and a shared agentic channel. Each account has a latent behavioural profile; an entity graph links accounts, devices, merchants and payees.

Hard negatives are the credibility test. I inject legitimate look-alikes (benign agentic shopping through trusted agents, large first-time payees, travel, collector accounts, recurring investments) into the exact feature regions where fraud lives.

0.83

best single-feature AUC

No one flag separates the classes, so the model must combine velocity, graph, behavioural and agent-identity signals. Without hard negatives, AUC would be a meaningless 1.0.

Agentic-commerce identity abuse - the 2026 frontier

A payment can now be initiated by an autonomous agent on a cardholder's behalf (Mastercard Agent Pay's Agentic Token; Visa's Trusted Agent Protocol). A hijacked or malicious agent spends inside someone else's mandate. It looks identical to a legitimate agent on velocity, device and cadence - only credential integrity separates them.

What I simulate

AGENT-HIJACK: one agent id replayed across many mandates, off-scope spend, weak attestation.

New defense family

Attestation · directory trust · mandate-cap breach · agent-id replay fan-out.

Unseen-vector result

15% → 100%

Trained with this attack fully removed, then scored on it. The supervised model alone vs the agent-identity features + novelty channel.

— THE CLOSED LOOP

The hardening curve - the whole thesis in one chart



Round 1: the red team finds evasion that collapses recall to **19%** - a static model would be blind. Retraining on those samples restores it to **83%**.

By the final round the attacker can no longer find easy evasion against the hardened detector. The loop converges because the defense generalised across the adversary's moves. A fixed detector does not adapt to evasion it has never seen; the closed loop trains on exactly those failures.

DEFEND

Strong, and reported the way a fraud desk reads it

1.0000

ROC-AUC

0.9999

PR-AUC

0.995

F1

100%

fraud value caught

140.4

alerts / 10k txns

0.000%

FPR @ 90% recall



Per-vector recall - deepfake authorised push (DF-APP) is hardest by design.

Two channels. LightGBM gradient boosting for known shapes, plus an unsupervised novelty channel (isolation forest + PCA) that flags attacks the model has never seen. ~45 features across event, velocity, graph and agent-identity families.

Explainable and cost-aware. Per-event reason codes via exact TreeSHAP; a cost-optimal operating point that minimises expected loss plus review workload.

CATCHING THE UNSEEN

Novelty channel - leave-one-vector-out

Each attack type is removed entirely from training, then scored. Higher = more of the unseen attack still caught. This is the direct answer to "novel, emerging".

unseen attack	supervised alone	+ novelty & agent-identity	blended
AGENT-CARD	99%	100%	100%
AGENT-HIJACK NEW VECTOR	15%	100%	100%
ATO-STUFF	58%	78%	80%
CARD-TEST	100%	96%	100%
DF-APP	80%	19%	32%
MULE-NET	100%	100%	100%
PIG-BUTCH	100%	37%	47%
STRUCT	100%	100%	100%
SYN-ID	2%	20%	20%

— VALIDATED BEYOND SYNTHETIC

It works on real fraud, and a GNN proves the graph matters

Same detector, real data

ROC 0.95 · PR 0.81

The unchanged two-channel ensemble on the ULB real-world card-fraud benchmark (284,807 genuine transactions, 492 fraud) - no dataset-specific tuning. The method transfers to real fraud.

The synthetic benchmark is genuinely hard: best single-feature AUC 0.83 (synthetic) vs 0.93 (real) - harder, not a toy.

GraphSAGE GNN on the transfer graph

**ring PR-AUC 0.84 →
0.998**

A two-layer GraphSAGE vs gradient boosting on the **same** per-account features. Message passing along the account graph lifts coordinated-ring detection by +0.15 PR-AUC, and holds under an inductive split with no test-node edges in training (+0.15, leakage-proof) - graph structure is decisive for mule rings, demonstrated not asserted.

EVALUATION RIGOR

The audits a fraud-ML reviewer would ask for

0.999

point-in-time PR-AUC

Structural features rebuilt with no look-ahead (cumulative up to each event).
Detection barely moves from 1.000 - not a leakage artifact.

± 0.003

recall across 5 seeds

ROC-AUC 1.00, recall 0.994. Nothing here is seed-dependent.

0.94 $\rightarrow$ 1.0

component ablation

Held-out PR-AUC event+velocity 0.94 to 1.000 once graph features are added - the graph is the decisive lift.

Competitive, but the loop is the point

On the real ULB benchmark, Chimera's static ensemble sits with standard models (PR-AUC: XGBoost 0.84, RandomForest 0.82, LightGBM 0.81, Chimera 0.78, LogReg 0.70). The contribution is not a better static classifier.

The loop transfers to real fraud

84% $\rightarrow$ 59% $\rightarrow$ 100%

Perturb real ULB fraud toward the legitimate distribution and recall drops; retrain on the evasive samples and it recovers. The methodology does not depend on the synthetic simulator.

Designed to sit where a bank already decides

Inline at authorisation. The detector scores the fields a switch already sees (ISO 8583 / UPI API) and returns a graded action: allow, step-up, hold, block. Velocity and graph features come from a streaming feature store, computed point-in-time-correct.

Native integration. Controls map to 3-DS step-up and RBI friction; graph features mirror NPCI MuleHunter; the agent-identity features model the public concepts Agent Pay and the Trusted Agent Protocol describe. Reason codes satisfy model-risk audit.

What paid infra buys

Streaming feature store (online features) · a temporal GNN for large coordinated rings · managed embeddings + vector DB for richer ideation grounding · a hosted low-latency endpoint for always-on variant generation. Quality dials, not a rewrite.

How this maps to the five judging criteria

Diversity of attacks	16 techniques, 9 simulated across card / real-time / agentic rails, plus a live agent that proposes more.
Fidelity of simulation	Latent-profile population, entity graph, hard negatives - best single-feature AUC only 0.83.
Detection efficacy	LightGBM + novelty + agent-identity features; 100% of fraud value caught at ~140.4 alerts / 10k.
Novelty	Closed loop with a measured hardening curve; agentic identity abuse; a live multi-agent red team.
Real-world feasibility	Auth-time schema, graded controls, NPCI-style graph signals, Agent Pay / TAP field mapping, streaming path.

Defensive by construction.

Chimera runs entirely on synthetic data it generates itself - no real cardholders, PII, credentials or live payment connectivity. The attack modules are parameterised statistical patterns that stress a detector, not operational playbooks or working exploit code. The red team exists only to harden the blue team.